

By Resolution on $C5, C3$ we get $C6: \square$

31. Use Resolution to show that the hypotheses "It is not raining or Yvette has her umbrella," "Yvette does not have her umbrella or she does not get wet," and "It is raining or Yvette does not get wet" imply that "Yvette does not get wet."

Soln: P : It is raining Q : Yvette has her umbrella
 R : Yvette gets wet

Premise: 1. $\neg P \vee Q$ 2. $\neg Q \vee \neg R$ 3. $P \vee \neg R$

Conclusion: $\neg R$

Clause $C1: \neg P \vee Q$ clause $C2: \neg Q \vee \neg R$ clause $C3: P \vee \neg R$

Clause $C4: \neg(\neg R) \equiv R$ (Double-negation Law)

By Resolution on clause $C3, C1$, we get $C5: \neg R \vee \neg Q$

By Resolution on clause $C2, C5$, we get $C6: \neg R$

By Resolution on clause $C4, C6$ we get: $\square$

32. Show that the equivalence $P \wedge \neg P \equiv F$ can be derived using resolution together with the fact that a conditional statement with a false hypothesis is True.

Soln: